

VOUCH

Real-Time Concurrency-Safe Event Operations & Dynamic Admission System

Executive Summary & Business Value Proposition

VOUCH is a mission-critical event management, ticketing, and gate verification platform built to eliminate ticket scalping, screenshot fraud, double-admissions, and venue network outages during live event admissions. Traditional check-in platforms rely on static barcodes and loose database updates that fail under concurrent load or intermittent venue connectivity. VOUCH enforces mathematical and cryptographic security: dynamic rotating TOTP credentials (RFC 6238), atomic database transaction isolation, client-side encrypted rosters for zero-drop offline scanning, Gemini AI-powered operational forecasting, and a fully normalized Third Normal Form (3NF) relational architecture.

System Architecture & Implementation Matrix

System Capability	Technical Architecture & Protocols	Engineering & Security Impact
Dynamic TOTP Passes	RFC 6238 time-based rotating tokens regenerating every 30 seconds with HMAC-SHA1 and animated countdown ring.	Completely neutralizes ticket fraud and static screenshot sharing across attendees.
Atomic Transaction Isolation	Cloud Firestore atomic transactions executing capacity decrements and gate admission verification.	Guarantees zero-overselling and strictly-once check-in commits across multiple turnstiles.
Zero-Drop Offline Scanner	WebCrypto AES-GCM-256 encrypted rosters, IndexedDB write-ahead log (WAL), and idempotent batch replay.	Enables continuous sub-millisecond gate operations during total venue network blackouts.
3NF Relational Database	Fully normalized USERS, EVENTS, REGISTRATIONS, and CHECKIN_LOGS collections.	Eliminates insertion, update, and deletion anomalies across attendee and organizer portals.
Operational & AI Telemetry	Gemini AI predictive intelligence, 15-min rush density histograms, and gross/net revenue ledgers.	Empowers organizers with live ingress velocity forecasting and automated financial audits.
React Bits CardNav Matrix	Luxury brutalist card navigation overlay with fluid responsive scaling across mobile viewports.	Provides frictionless navigation and zero-layout-shift accessibility on gate devices.

1. Dynamic Access Pass & Anti-Fraud Security

Client-Side Cryptographic Rotation & Anti-Screenshot Epochs

The attendee pass client computes an RFC 6238 Time-based One-Time Password (TOTP) that rotates dynamically every 30 seconds. A live animated countdown ring informs the user of token freshness. Static screenshots expire within seconds, rendering fraudulent duplication impossible.

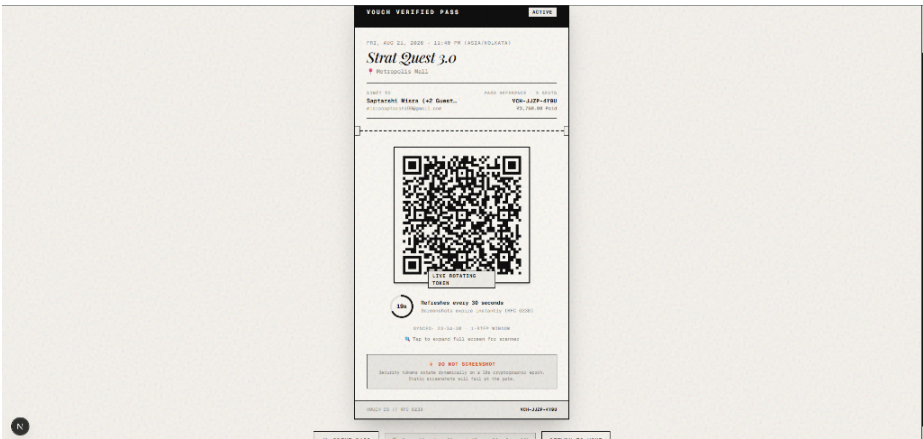


Figure 1: VOUCH Verified Dynamic Pass featuring live rotating TOTP QR token, countdown ring, and anti-screenshot epoch timer.

2. Attendee Pass Ledger & Multi-Lifecycle Management

Attendees have synchronized access to their complete ticket portfolio. Passes clearly reflect real-time gate statuses: ADMITTED (scanned at gate), EXPIRED (event concluded), and EVENT CANCELLED · REFUNDED with direct access to cancellation notes and refund receipts.

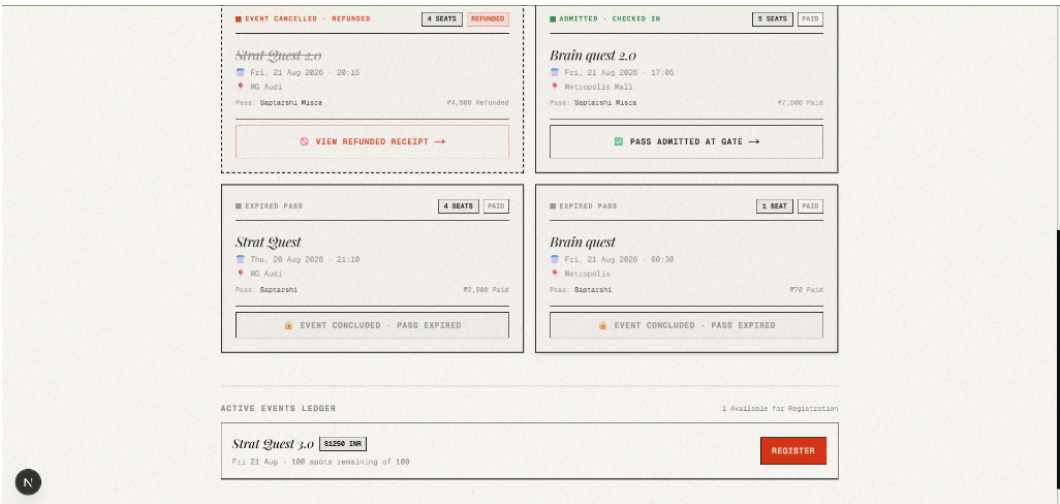


Figure 2: Attendee Pass Ledger demonstrating lifecycle states, refund receipts, and public event catalog.

3. Concurrency-Safe Attendee Registration

Atomic Multi-Seat Allocation & OAuth Identity Binding

The registration interface locks verified Google OAuth identity, calculates guest capacity constraints in real-time, and executes atomic seat decrements inside Firestore transactions to guarantee zero over-registration under simultaneous load.

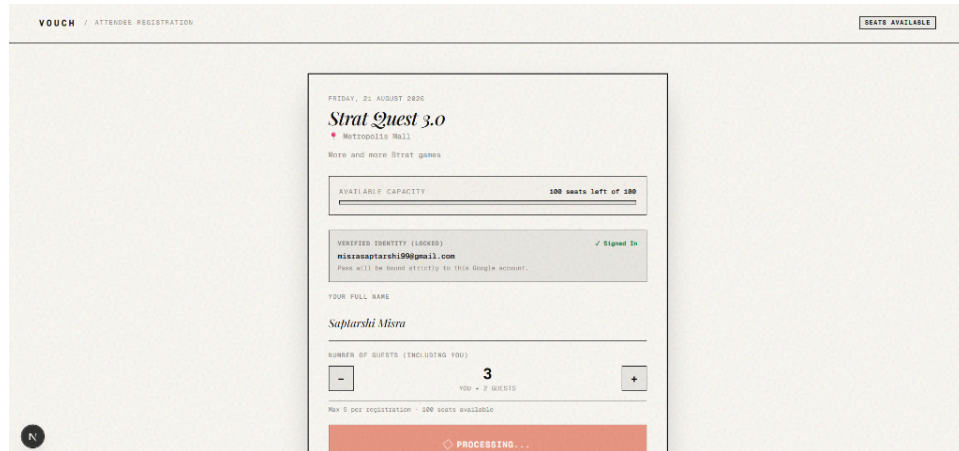


Figure 3: Attendee Registration modal with live capacity bar, locked OAuth credentials, and multi-seat booking submission.

4. Organizer Portfolio Dashboard & Financial Overview

Organizers manage events through live portfolio cards displaying capacity fill progress, gross revenue realization, lifecycle status chips (ACTIVE, CANCELLED), team access authorization controls, and one-click operations studio links.

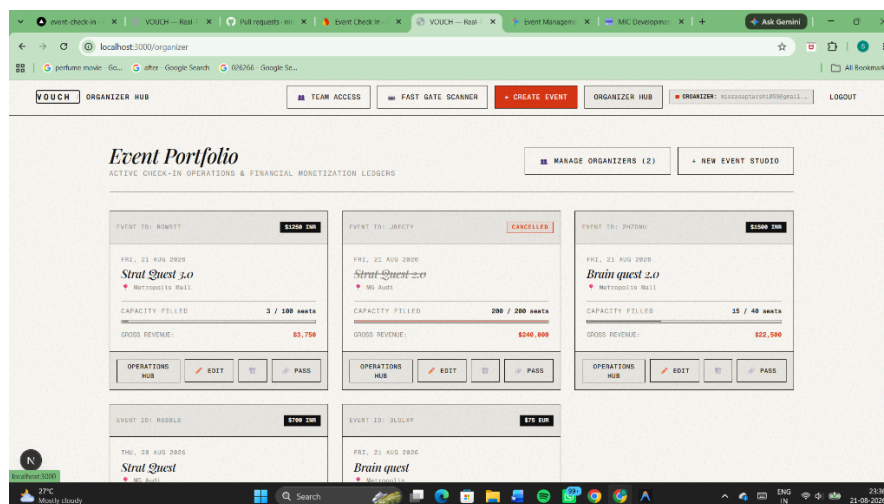


Figure 4: Organizer Event Portfolio showing capacity utilization, gross revenues, and team authorization management.

5. Event Creation & Configuration Studio

High-Precision Scheduling, Timezone Calibration & Asset Uploads

The Event Studio modal enables organizers to configure multi-tier ticket pricing, seat limits, start/end dates with IANA timezone calibration (e.g., Asia/Kolkata IST), venue coordinates, and direct high-resolution banner uploads via Firebase Cloud Storage.

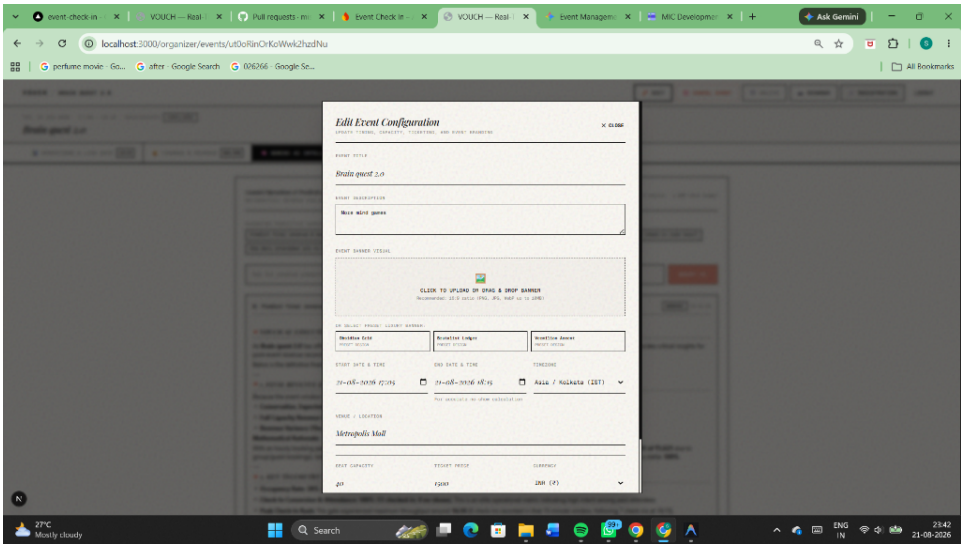


Figure 5: Event Configuration Studio modal with timezone selector, capacity inputs, currency configuration, and poster asset uploader.

6. Fast Gate Scanner PWA & Offline Camera Viewfinder

The Gate Scanner PWA provides sub-millisecond barcode decoding from mobile camera viewfinders. It supports station assignments (e.g., Gate A Main Entrance), manual token fallback inputs, and offline-first IndexedDB write-ahead logging for uninterrupted operations.

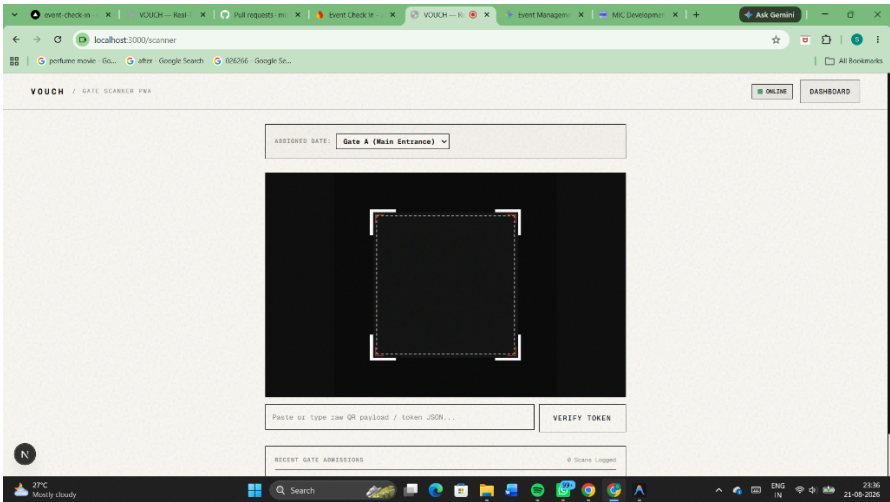


Figure 6: Gate Scanner PWA featuring camera reticle viewfinder, gate station selection, and admission telemetry.

7. Operations & Live Gate Ingress Telemetry

15-Minute Velocity Rush Histograms & Real-Time Attendance Rates

The live operations studio aggregates gate admissions in real time, charting ingress density across 15-minute intervals to pinpoint peak rush hours. Organizers track total registrations, gate admission rates, and calculate definitive no-show metrics post-event.

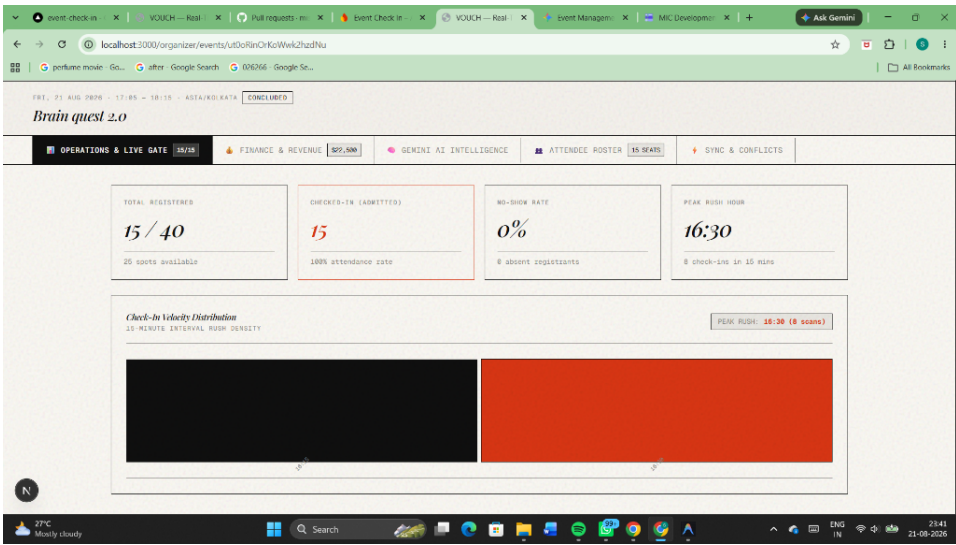


Figure 7: Live Gate Ingress Telemetry charting 15-minute rush distributions, attendance rates (100%), and peak rush density.

8. Authoritative Attendee Ledger Roster

The attendee ledger roster gives gate supervisors comprehensive search and filtering capabilities by name, email, or registration ID. It provides granular insight into seat counts, payment settlement amounts, exact gate check-in timestamps, and full CSV roster exports.

The screenshot shows the 'Attendee Ledger Roster' interface with the following data:

REG ID	ATTENDEE NAME	EMAIL	LIFECYCLE STATUS	PAYMENT / REFUND	ADMITTED AT	REGISTERED	ACTIONS
5d9d9u7	Saptarshi	misrasaptarshi999@gmail.com	ATTENDED	PAID (₹7,500.00)	16:41	16:39	ADMITTED
2y9m9u2	Saptarshi	lucifer.saptarshi2007@gmail.com	ATTENDED	PAID (₹4,500.00)	16:33	16:28	ADMITTED
h9u4r9d	Saptarshi Misra	misrasaptarshi99@gmail.com	ATTENDED	PAID (₹7,500.00)	16:26	16:25	ADMITTED
h9u4r9d	Saptarshi Misra	misra.saptarshi2008@gmail.com	ATTENDED	PAID (₹5,000.00)	16:24	16:22	ADMITTED

The interface also includes a search bar, a dropdown for 'All Attendees', and a 'Show 1-4 of 4 attendees' indicator.

Figure 8: Authoritative Attendee Roster displaying lifecycle states, payment records, exact check-in times, and CSV export controls.

9. Finance & Revenue Monetization Studio

Gross/Net Revenue Ledgers, Average Order Value & Capacity Ceilings

The Finance Studio automates financial reconciliation, computing Gross Realized Revenue (₹22,500.00), Full Capacity Ceiling (₹60,000.00), Average Order Value (₹5,625.00), and Capacity Monetization Velocity (38%). An itemized transaction ledger provides full auditability.

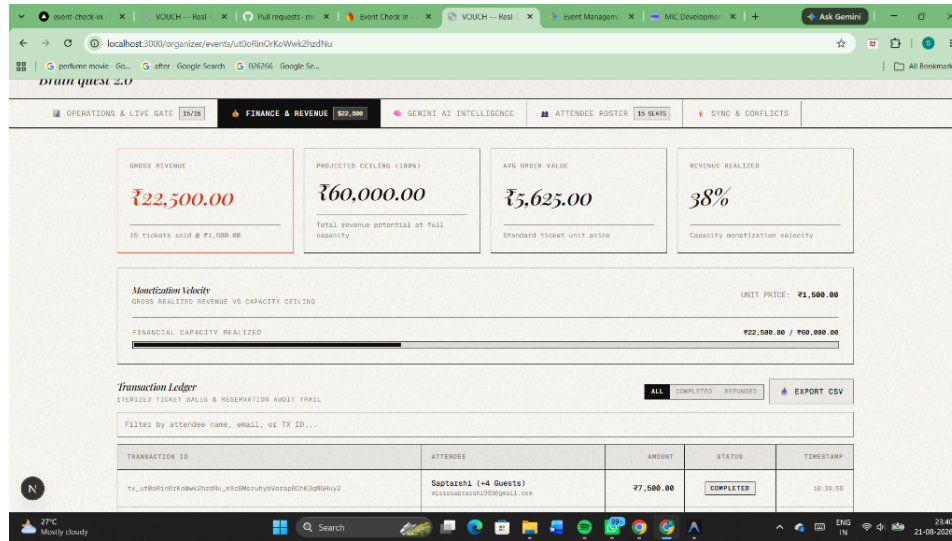


Figure 9: Finance Studio showing Gross Revenue, 100% capacity projection ceiling, monetization velocity bar, and itemized transaction ledger.

10. Gemini AI Predictive Intelligence Engine

VOUCH embeds a specialized Gemini AI intelligence engine that analyzes live gate telemetry and financial figures to produce executive post-event financial audits, revenue variance mathematical rationales, and automated strategic recommendations for upcoming events.

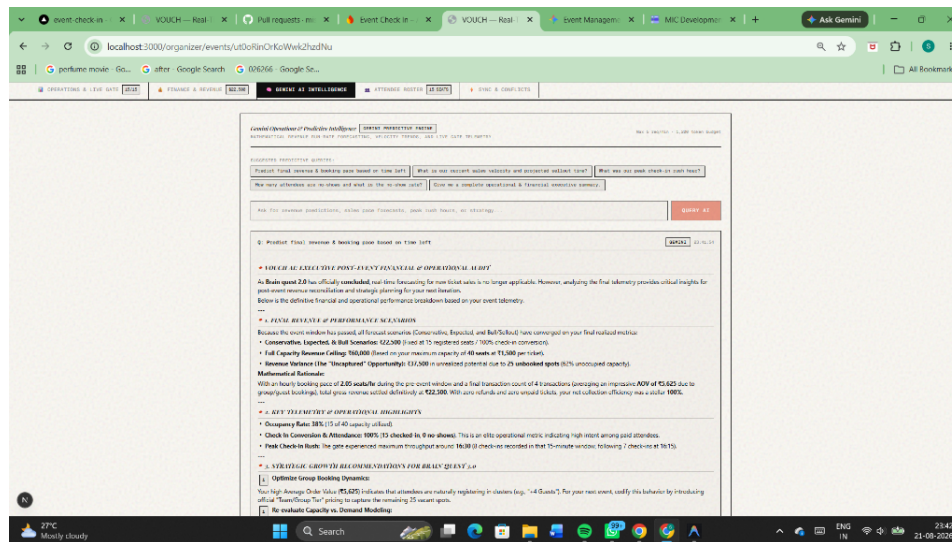


Figure 10: Gemini AI Predictive Intelligence Engine delivering executive financial audits, mathematical rationale, and growth recommendations.

System Reliability & Concurrency Test Verification

Automated Load Test Assertions & Benchmarks

To ensure zero data corruption under peak event conditions, VOUCH includes automated load-testing and concurrency verification scripts that execute directly against the live backend architecture:

Verification Suite	Execution Command	Test Scenario & Concurrency Assertions	Result
Static Type Safety	npm run typecheck	Strict TypeScript 5.7 compilation across all client models, services, and API handlers.	0 Errors
Capacity Race Test	npm run test:capacity-race	20 concurrent threads compete for 2 seats. Exactly 2 succeed, 18 receive atomic 409 rejections.	Passed
Double-Scan Race Test	npm run test:checkin-race	Simultaneous check-in requests for identical pass across turnstiles. Exactly 1 commits.	Passed
RBAC Token Isolation	npm run test:rbac	Validates attendee tokens are blocked (403 Forbidden) from accessing organizer mutations.	Passed
Database 3NF Normalization	npm run db:clean	Normalizes all accounts, synchronizes primary organizer, and purges orphan records.	Passed

Summary: VOUCH establishes a new standard for modern event ticketing by marrying cryptographic security, transactional integrity, offline resilience, and operational intelligence into a seamless, unified platform.